1 $5\sqrt{3} - 3\sqrt{3} + 2\sqrt{3}$ **[M1]**

$4\sqrt{3}$ **[A1]**

[Total Marks: 2]

2 $5\sqrt{3}$ or $4\sqrt{3}$ or $3\sqrt{3}$ seen **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 2]

3 $6\sqrt{2}$ or $5\sqrt{2}$ or $3\sqrt{2}$ seen **[M1]**

$8\sqrt{2}$ **[A1]**

[Total Marks: 2]

4 $5\sqrt{3}$ or $3\sqrt{3}$ or $2\sqrt{3}$ seen **[M1]**

$4\sqrt{3}$ **[A1]**

[Total Marks: 2]

5 $\sqrt{25 \times 3} + \sqrt{16 \times 3} - \sqrt{9 \times 3} = 5\sqrt{3} + 4\sqrt{3} - 3\sqrt{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 2]

6 (a) $3\sqrt{5}$ **[B1]**

(b) $2\sqrt{5}$ or $4\sqrt{5}$ seen **[M1]**

$6\sqrt{5}$ **[A1]**

[Total Marks: 3]

7 $\sqrt{75} = 5\sqrt{3}$ **[B1]**

$\sqrt{27} = 3\sqrt{3}$ or $\sqrt{12} = 2\sqrt{3}$ **[B1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 3]

8 $\sqrt{48} = 4\sqrt{3}$ **[B1]**

$\sqrt{12} = 2\sqrt{3}$ **[B1]**

$4\sqrt{3} - 2\sqrt{3} = 2\sqrt{3}$ **[B1]**

[Total Marks: 3]

9 **(a)** M1 for $5\sqrt{3}$ or $2\sqrt{3}$

$7\sqrt{3}$ **[A1]**

(b) M1 for $\dfrac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ or $\dfrac{14\sqrt{7}}{7}$

$2\sqrt{7}$ **[A1]**

[Total Marks: 4]

10 **(a)** $\sqrt{25 \times 3}$ or $\sqrt{25}\sqrt{3}$ **[M1]**

$5\sqrt{3}$ **[A1]**

(b) $6\sqrt{2} - 2\sqrt{2}$ **[M1]**

$4\sqrt{2}$ **[A1]**

[Total Marks: 4]

11 **(a)** **(i)** $3\sqrt{5} + 2\sqrt{5}$ **[M1]**

$5\sqrt{5}$ **[A1]**

(ii) Multiplies numerator and denominator by $\sqrt{3}$: $\dfrac{18\sqrt{3}}{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 4]

12 **(a)** $3\sqrt{5}$ **[B1]**

(b) $6\sqrt{2} + 5\sqrt{2} - 3\sqrt{2}$ **[M1]**

$8\sqrt{2}$ **[A1]**

(c) $\dfrac{21\sqrt{7}}{\sqrt{7}\sqrt{7}}$ or $\dfrac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Total Marks: 5]

13 **(a)** $8 + 4\sqrt{3} - 2\sqrt{3} - 3$ **[M1]**

$5 + 2\sqrt{3}$ **[A1]**

(b) $\dfrac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Total Marks: 4]

14 **(a)** $9 - 3\sqrt{5} - 3\sqrt{5} + 5$ (or $9 - 6\sqrt{5} + 5$) **[M1]**

$14 - 6\sqrt{5}$ **[A1]**

(b) $\dfrac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 4]

15 (a) $12 + 3\sqrt{5} - 8\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) Multiplies numerator and denominator by $\sqrt{7}$ to get $\dfrac{14\sqrt{7}}{7}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 4]

16 (a) (a) $12 + 3\sqrt{5} - 8\sqrt{5} - 2(5)$ or $12 - 5\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) $\dfrac{15 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$ or $\dfrac{15\sqrt{3}}{3}$ **[M1]**

$5\sqrt{3}$ **[A1]**

[Total Marks: 4]

17 (a) $12 - 3\sqrt{2} + 4\sqrt{2} - 2$ **[M1]**

$10 + \sqrt{2}$ **[A1]**

(b) Multiply numerator and denominator by $(5 + \sqrt{3})$ **[M1]**

Denominator evaluates to $25 - 3 = 22$ **[M1]**

$5 + \sqrt{3}$ **[A1]**

[Total Marks: 5]

18 (a) M1 for $9 + 3\sqrt{2} + 3\sqrt{2} + 2$ (allow one numerical error in expansion)

$11 + 6\sqrt{2}$ **[A1]**

(b) M1 for multiplying numerator and denominator by $(3 + \sqrt{5})$

M1 for denominator evaluated to 4 (from $3^2 - (\sqrt{5})^2$)

$3 + \sqrt{5}$ **[A1]**

[Total Marks: 5]

19 (a) $20 - 9$ or $4(5) - 9$ **[M1]**

11 **[A1]**

(b) Multiplies numerator and denominator by $(3+\sqrt{2})$ **[M1]**

Denominator of $9-2$ or 7 seen **[M1]**

$6+2\sqrt{2}$ **[A1]**

[Total Marks: 5]

20 (a) $12+15\sqrt{2}-2\sqrt{2}-5$ **[M1]**

$7+13\sqrt{2}$ **[A1]**

(b) Multiplies numerator and denominator by $(4+\sqrt{3})$ **[M1]**

Denominator of $16-3$ or 13 **[M1]**

$8+2\sqrt{3}$ or $2(4+\sqrt{3})$ **[A1]**

[Total Marks: 5]

21 (a) $\dfrac{15\sqrt{5}}{5}$ **[M1]**

$3\sqrt{5}$ **[A1]**

(b) $10-2\sqrt{3}+5\sqrt{3}-3$ **[M1]**

$7+3\sqrt{3}$ **[A2]**

[Total Marks: 5]

22 (a) $6-3\sqrt{5}+2\sqrt{5}-5$ **[M1]**

$1-\sqrt{5}$ **[A1]**

(b) $\dfrac{14\times\sqrt{7}}{\sqrt{7}\times\sqrt{7}}$ **[M1]**

$\dfrac{14\sqrt{7}}{7}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 5]

23 (a) $15+5\sqrt{7}-6\sqrt{7}-2(7)$ **[M1]**

$1-\sqrt{7}$ **[A1]**

(b) Multiplies numerator and denominator by $(5+\sqrt{3})$: $\dfrac{22(5+\sqrt{3})}{(5-\sqrt{3})(5+\sqrt{3})}$ **[M1]**

Denominator becomes $25-3=22$ **[M1]**

$5+\sqrt{3}$ **[A1]**

[Total Marks: 5]

24 **(a)** $20 + 8\sqrt{2} - 15\sqrt{2} - 6(2)$ **[M1]**

$8 - 7\sqrt{2}$ **[A1]**

(b) $\dfrac{11(4+\sqrt{5})}{(4-\sqrt{5})(4+\sqrt{5})}$ **[M1]**

$\dfrac{11(4+\sqrt{5})}{16-5}$ or $\dfrac{11(4+\sqrt{5})}{11}$ **[M1]**

$4 + \sqrt{5}$ **[A1]**

(c) $3x\sqrt{3} - x\sqrt{3} = 18$ **[M1]**

$2x\sqrt{3} = 18$ or $x\sqrt{3} = 9$ **[M1]**

$x = \dfrac{9}{\sqrt{3}} = 3\sqrt{3}$ **[A1]**

[Total Marks: 8]

25 $\times \dfrac{3+\sqrt{7}}{3+\sqrt{7}}$ **[M1]**

Numerator $15 + 5\sqrt{7} + 3\sqrt{7} + 7$ or $22 + 8\sqrt{7}$ **[B1]**

Denominator $9 - 7$ or 2 **[B1]**

$11 + 4\sqrt{7}$ **[A1]**

[Total Marks: 4]

26 M1 for rearranging to group x terms, e.g., $x\sqrt{3} - x = 2 + \sqrt{12}$

M1 for factorising x, e.g., $x(\sqrt{3} - 1) = 2 + 2\sqrt{3}$

M1 for multiplying by the conjugate to rationalise: $\dfrac{2+2\sqrt{3}}{\sqrt{3}-1} \times \dfrac{\sqrt{3}+1}{\sqrt{3}+1}$

M1 for correctly expanding the numerator to $8 + 4\sqrt{3}$ or evaluating the denominator to 2

$4 + 2\sqrt{3}$ **[A1]**

[Total Marks: 5]

27 Forms an equation for the area: $(a + b\sqrt{3})(2 - \sqrt{3}) = 1 + 4\sqrt{3}$ **[M1]**

Expands the brackets: $2a - a\sqrt{3} + 2b\sqrt{3} - 3b = 1 + 4\sqrt{3}$ **[M1]**

Equates rational and irrational parts to form simultaneous equations: $2a - 3b = 1$ and $-a + 2b = 4$ **[M1]**

Attempts to solve the simultaneous equations (e.g. multiplies second equation by 2: $-2a + 4b = 8$ and adds to first to get $b = 9$) **[M1]**

$a = 14$, $b = 9$ **[A1]**

[Total Marks: 5]

28 **(a)** Multiplies numerator and denominator by $(2+\sqrt{3})$ **[M1]**

Numerator expands to $10+5\sqrt{3}+2\sqrt{3}+3$ **[M1]**

Denominator is $4-3=1$ leading to final answer $13+7\sqrt{3}$ **[A1]**

(b) $x=\dfrac{14}{3-\sqrt{2}}$ **[M1]**

Multiplies numerator and denominator by $(3+\sqrt{2})$ to get $\dfrac{14(3+\sqrt{2})}{9-2}$ **[M1]**

$x=6+2\sqrt{2}$ **[A1]**

[Total Marks: 6]

29 **(a)** Multiply numerator and denominator by $2+\sqrt{3}$ **[M1]**

Numerator becomes $(5+\sqrt{3})(2+\sqrt{3})=10+5\sqrt{3}+2\sqrt{3}+3$ (giving $13+7\sqrt{3}$) **[M1]**

$13+7\sqrt{3}$ **[A1]**

(b) Length $=\dfrac{12+4\sqrt{5}}{3+\sqrt{5}}$ **[M1]**

Multiply numerator and denominator by $3-\sqrt{5}$ **[M1]**

Numerator becomes $(12+4\sqrt{5})(3-\sqrt{5})=36-12\sqrt{5}+12\sqrt{5}-20=16$ **[M1]**

4 **[A1]**

[Total Marks: 7]

30 **(a)** Multiplies numerator and denominator by $(5+\sqrt{3})$ **[M1]**

Denominator evaluates to $25-3$ or 22 **[M1]**

$5+\sqrt{3}$ **[A1]**

(b) $x(\sqrt{2}-1)=\sqrt{8}$ or $x(\sqrt{2}-1)=2\sqrt{2}$ **[M1]**

$x=\dfrac{2\sqrt{2}}{\sqrt{2}-1}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{2}+1)$ **[M1]**

$4+2\sqrt{2}$ **[A1]**

[Total Marks: 7]

31 **(a)** $\dfrac{1}{2}\times(2+\sqrt{3})\times h=5+3\sqrt{3}$ **[M1]**

$h=\dfrac{10+6\sqrt{3}}{2+\sqrt{3}}$ **[M1]**

Multiply numerator and denominator by $(2-\sqrt{3})$ **[M1]**

$h=\dfrac{20-10\sqrt{3}+12\sqrt{3}-18}{4-3}=2+2\sqrt{3}$ **[A1]**

(b) $y^2=(2+\sqrt{3})^2+(2+2\sqrt{3})^2$ **[M1]**

$y^2 = (4 + 4\sqrt{3} + 3) + (4 + 8\sqrt{3} + 12)$ **[M1]**

$23 + 12\sqrt{3}$ **[A1]**

[Total Marks: 7]

32 **(a)** $\dfrac{5 + 2\sqrt{3}}{2 + \sqrt{3}}$ **[M1]**

Multiplies numerator and denominator by $(2 - \sqrt{3})$ **[M1]**

$4 - \sqrt{3}$ **[A1]**

(b) $x\sqrt{5} - 2x = 2 + \sqrt{20}$ or $x(\sqrt{5} - 2) = \ldots$ **[M1]**

$x = \dfrac{2 + 2\sqrt{5}}{\sqrt{5} - 2}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{5} + 2)$ **[M1]**

$14 + 6\sqrt{5}$ **[A1]**

[Total Marks: 7]

33 **(a)** $\dfrac{14 - 2\sqrt{5}}{3 + \sqrt{5}}$ **[M1]**

Multiplies numerator and denominator by $(3 - \sqrt{5})$ **[M1]**

Numerator expands to $42 - 14\sqrt{5} - 6\sqrt{5} + 10$ (or $52 - 20\sqrt{5}$) **[M1]**

$13 - 5\sqrt{5}$ **[A1]**

(b) Uses Pythagoras' theorem: $(3 + \sqrt{5})^2 + (13 - 5\sqrt{5})^2$ **[M1]**

$14 + 6\sqrt{5}$ or $294 - 130\sqrt{5}$ seen **[M1]**

$308 - 124\sqrt{5}$ with no errors seen **[A1]**

[Total Marks: 7]

34 **(a)** Multiplies numerator and denominator by $(3 + \sqrt{3})$: $\dfrac{6(3 + \sqrt{3})}{(3 - \sqrt{3})(3 + \sqrt{3})}$ **[M1]**

Expands denominator correctly: $9 - 3$ or 6 **[M1]**

$\dfrac{6(3 + \sqrt{3})}{6}$ **[M1]**

$3 + \sqrt{3}$ **[A1]**

(b) $\dfrac{2\sqrt{6} + 3\sqrt{6}}{\sqrt{3}}$ or $\dfrac{5\sqrt{6}}{\sqrt{3}}$ or $5\sqrt{2}$ **[M1]**

$\dfrac{10\sqrt{2}}{2}$ or $5\sqrt{2}$ **[M1]**

0 (or $0\sqrt{2} + 0\sqrt{3}$) **[A1]**

[Total Marks: 7]

35 **(a)** Area $= \frac{1}{2} \times \text{base} \times \text{height} \implies 5 + 2\sqrt{2} = \frac{1}{2} \times (3 + \sqrt{2}) \times h$ **[M1]**

$h = \frac{2(5 + 2\sqrt{2})}{3 + \sqrt{2}}$ **[A1]**

(b) Multiply numerator and denominator by $(3 - \sqrt{2})$: $\frac{2(5 + 2\sqrt{2})(3 - \sqrt{2})}{(3 + \sqrt{2})(3 - \sqrt{2})}$ **[M1]**

Denominator: $3^2 - (\sqrt{2})^2 = 9 - 2 = 7$ **[B1]**

Numerator: $2(15 - 5\sqrt{2} + 6\sqrt{2} - 4) = 2(11 + \sqrt{2}) = 22 + 2\sqrt{2}$ **[M1]**

Height $= \frac{22 + 2\sqrt{2}}{7}$ **[A1]**

(c) $\frac{1(\sqrt{7} + 2)}{(\sqrt{7} - 2)(\sqrt{7} + 2)}$ **[M1]**

$\frac{\sqrt{7} + 2}{7 - 4} = \frac{\sqrt{7} + 2}{3}$ **[A1]**

[Total Marks: 8]

36 **(a)** $\frac{1}{2} \times (\sqrt{6} - \sqrt{2})(\sqrt{6} + \sqrt{2})$ **[M1]**

$\frac{1}{2}(6 - 2)$ **[M1]**

2 **[A1]**

(b) $(\text{Hypotenuse})^2 = (\sqrt{6} - \sqrt{2})^2 + (\sqrt{6} + \sqrt{2})^2$ **[M1]**

$(6 - 2\sqrt{12} + 2) + (6 + 2\sqrt{12} + 2)$ or $8 - 2\sqrt{12} + 8 + 2\sqrt{12}$ **[M1]**

$(\text{Hypotenuse})^2 = 16$ **[A1]**

$\text{Hypotenuse} = \sqrt{16} = 4$ **[A1]**

(c) Area scale factor $= \frac{18}{2} = 9$ **[M1]**

Length scale factor $= \sqrt{9} = 3$ **[M1]**

$4 \times 3 = 12$ **[A1]**

[Total Marks: 10]